

Macro Sign Service

Automated VBA Macro Signing for Enterprise

Presented to: CISO | UK Macro Signing Team | Switzerland Team

March 2026

One service. Two entities. Zero manual effort.

Agenda

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The Problem

Current state & pain points of manual macro signing

02

The Solution

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The Problem -Current Manual Workflow

How macro signing works today (UK)

CURRENT PROCESS

- 1 Developer uploads .vba file to ServiceNow
- 2 SNOW ticket created and assigned to UK signing team
- 3 Team member manually reviews the macro
- 4 Team member manually signs with certificate tooling
- 5 Team member uploads signed file back to SNOW
- 6 Developer downloads signed file -hours to days later

Switzerland (CH): No macro signing solution exists -macros are unsigned or risk-accepted

The Problem -Bottlenecks & Costs

Bottleneck	Impact
Manual handoff	Developer uploads -> waits -> team picks up -> signs -> returns
Time delay	Hours to days turnaround per signing request
Human error	Wrong certificate, missed files, copy-paste mistakes
No audit trail	Limited traceability of who signed what and when
Scaling cost	Dedicated team members tied up with repetitive tasks
No CI/CD integration	Macros in repos remain unsigned until manual process completes
Single point of failure	Key team members on leave = signing queue stalls

Result: Slow, error-prone, costly, unscalable, no CH coverage

The Solution -Macro Sign Service

An enterprise-grade, automated macro signing service that eliminates manual signing workflows entirely.

ServiceNow (Synchronous)

Submit macro via SNOW form

Signed content returned instantly

Single HTTP call, 200 OK response

Sub-second turnaround

CI/CD Pipeline (Asynchronous)

Auto-sign on git push / PR merge

GitHub Actions, Azure DevOps, Jenkins

202 Accepted -> poll -> completed

Webhook notifications supported

BEFORE

Upload -> Wait hours/days -> Manual sign -> Return

AFTER

Upload -> Signed instantly | git push -> Signed automatically

Zero human intervention. 24/7 availability. Full audit trail.

Value Stream & Benefits

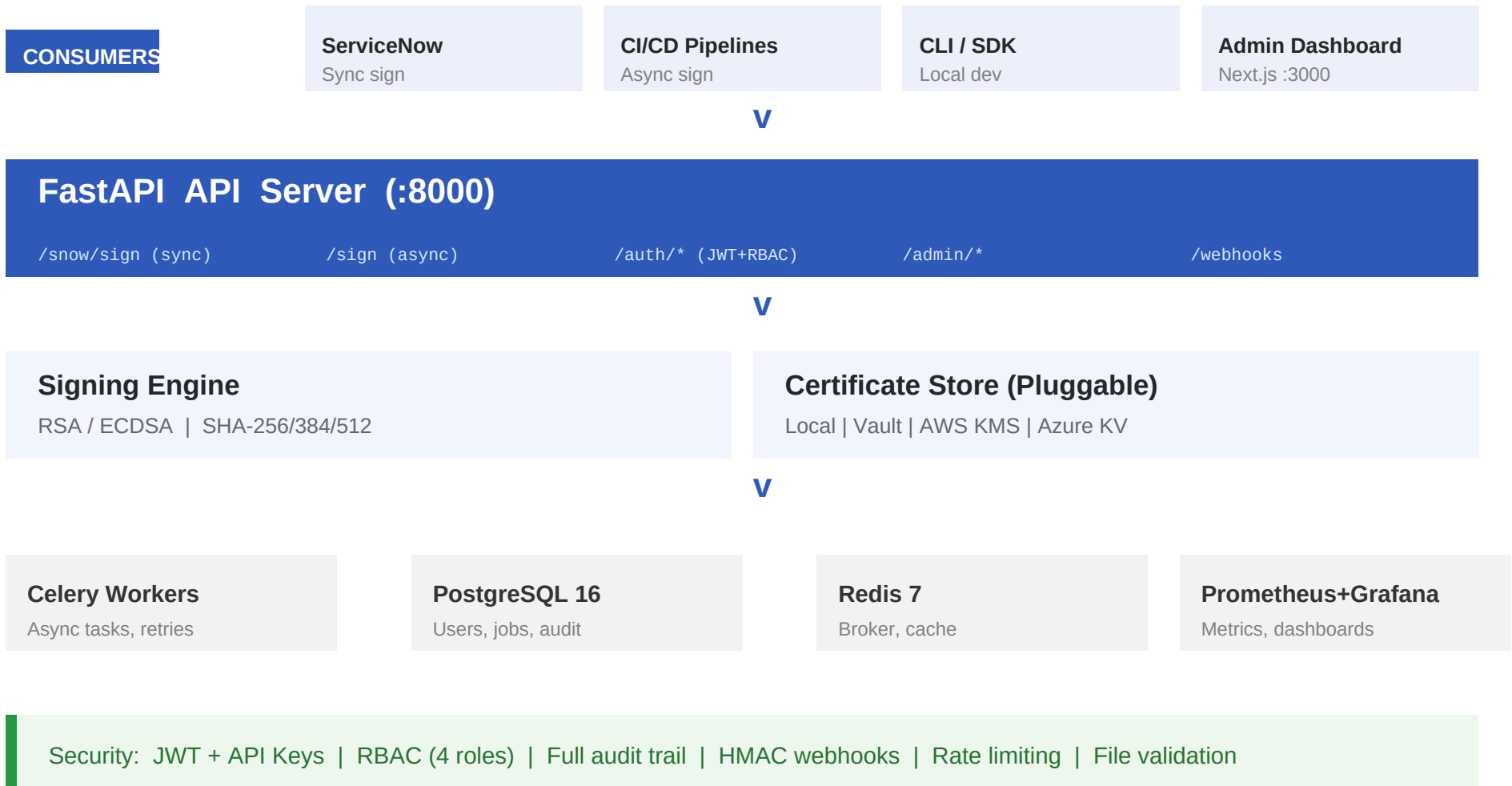
Quantified impact

Metric	Before (Manual)	After (Automated)	Improvement
Turnaround time	Hours to days	Sub-second	~99.9% faster
Human effort	15-30 min / signing	0 min	100% eliminated
Error rate	Manual mistakes	Deterministic	Near-zero
Audit completeness	Partial / manual	Every op logged	100% traceability
Availability	Business hours only	24/7 automated	Always-on
CH coverage	None	Full capability	New capability

STRATEGIC BENEFITS

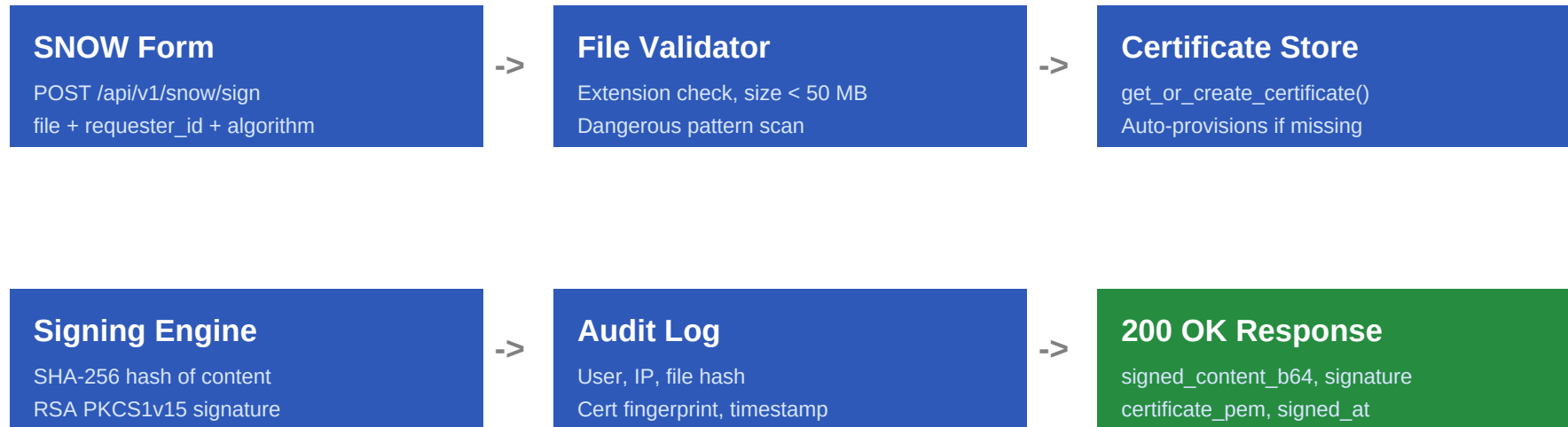
- Security posture -every macro cryptographically signed; full compliance audit trail
- Developer velocity -no more waiting; signing in CI/CD or instant via SNOW
- Operational cost -frees signing team for higher-value security work
- Scalability -hundreds of requests/min; no human bottleneck
- Group standardization -one service for UK + CH (+ future entities)
- Risk reduction -eliminates unsigned macros in CH; reduces human error in UK

Architecture -System Overview



Architecture -ServiceNow Signing Flow

Synchronous -single HTTP call, instant response



Response (200 OK):

```
"status": "signed"      "signature": "<hex>"      "file_hash": "<hex>"
"certificate_fingerprint": "AB:CD:..."  "certificate_pem": "-----BEGIN..."  "signed_at": "..."
```


Technology Stack & Security Controls

TECH STACK

Layer	Technology
API	Python 3.11+, FastAPI, Uvicorn
Task Queue	Celery + Redis broker
Database	PostgreSQL 16, SQLAlchemy (async)
Crypto	RSA, ECDSA, X.509, SHA-256/384/512
Auth	JWT (30 min) + API Keys + RBAC
Dashboard	Next.js 14, React 18, Tailwind CSS
Cert Backends	Local Vault AWS KMS Azure KV
Monitoring	Prometheus + Grafana
Deployment	Docker Compose (dev), K8s + Helm (prod)

SECURITY

- Private keys never leave Vault / KMS
- JWT tokens (30 min expiry) + refresh
- API keys with mss_ prefix (secrets scan)
- RBAC: Admin, Manager, Developer, Viewer
- Full audit: user, IP, hash, cert, timestamp
- File validation: ext, size, pattern scan
- HMAC-SHA256 signed webhook payloads
- Rate limiting: 60 req/min per user
- Bcrypt password hashing (cost 12)

Live Demo

What we will demonstrate:

- | | | |
|---|-------------------------------|--|
| 1 | Service Startup | <code>docker-compose up</code> -all services come online |
| 2 | Health Check | <code>curl /api/v1/health</code> -confirms DB + Redis connectivity |
| 3 | SNOW Sync Signing | <code>POST .vba to /snow/sign</code> -signed content back instantly (200 OK) |
| 4 | Signature Verification | <code>POST file + signature to /snow/verify</code> -confirms <code>is_valid: true</code> |
| 5 | CI/CD Async Signing | <code>POST /sign -> job_id -> poll /status/{job_id} -> completed</code> |
| 6 | Admin Dashboard | Web UI: signing jobs, audit logs, certificates, users/teams |
| 7 | Audit Trail | Full log: who signed, when, file hash, which certificate |
| 8 | Certificate Mgmt | List certs, view details, auto-provisioning |

```
# Sign a macro -instant response
curl -X POST http://localhost:8000/api/v1/snow/sign -F "file=@macro.vba"
```

Next Steps -Entity Coverage

Entity	Current State	With Macro Sign Service
UK	Manual SNOW workflow	Automated signing via SNOW + CI/CD
Switzerland (CH)	No solution exists	Full signing capability -same standards
Future entities	No standardization	Onboard to the same group service

This is not just a UK tool - it is a Group Service opportunity.

KEY STAKEHOLDERS

CISO	Approve approach, sponsor Group Service elevation
UK Signing Team	Primary users, validate workflow replacement
CH Team	New capability beneficiaries, validate requirements
Group InfoSec	Co-own the service, define certificate & signing policies
Group IAM	Integrate with centralised identity (SSO, API auth)
Platform / Infra	Kubernetes hosting, monitoring, SLA

Roadmap -PoC to Group Service

PHASE 1 -NOW

PoC / Demo

- Demo to CISO & teams
- Validate architecture & security
- Gather feedback from UK + CH
- Confirm fit for both entities



PHASE 2

Production Pilot

- Deploy to Kubernetes (prod)
- Integrate with Group IAM / SSO
- Enterprise CA certificates
- Onboard UK + CH teams
- Connect to SNOW production



PHASE 3

Group Service

- Official Group Service status
- Co-owned: Group InfoSec + IAM
- HSM / PKCS#11 for hardware keys
- SLA-backed availability
- Onboard future entities

When elevated to Group Service: convert PoC to production service with Group InfoSec & Group IAM ownership

Summary

	Manual (Today)	Macro Sign Service
Speed	Hours to days	Sub-second
Cost	Dedicated team effort	Zero marginal cost
Coverage	UK only	UK + CH + future
Audit	Partial	Complete, automated
CI/CD	Not integrated	Native integration
Availability	Business hours	24/7
Risk	Human error + no CH	Deterministic, enforced

One service. Two entities. Zero manual effort.

Thank you

Questions?